

Wireless Power Transmission

Section 1 — The Technology and the Vision

Wireless power transmission delivers usable electrical energy across open space, with no wire, plug, or contact between source and receiver. The physics has been settled since 1975, when Brown and JPL beamed 30 kW across 1.54 km at Goldstone and recovered it as DC at over 80 percent conversion efficiency. Fifty years later that demonstration has never been meaningfully surpassed. What has changed is not whether it works. It is whether it can become infrastructure.

The technology splits by range. Near-field inductive coupling — the Qi charger under every phone — works to a few centimetres. Far-field radiative transfer, where energy leaves as a propagating wave and is captured by a rectenna, is unbounded in range and is the only regime that supports a genuine infrastructure vision. Everything else is a better cable.

Terrestrial: rooms and buildings carry ambient transmitters the way they carry lighting and Wi-Fi. Power fills the volume rather than targeting a device. Anything with an antenna and a rectifier harvests continuously on entry, with no pairing and no negotiation. Adding a receiver costs the infrastructure nothing — the system scales like broadcast, not like a distribution grid.

Orbital: a collector in geostationary orbit sits in sunlight over 99 percent of the year, receiving 1,361 W/m² undiminished by atmosphere, against a terrestrial capacity factor of 15–25 percent. The energy converts to microwave, beams to a ground rectenna, and rectifies back to grid power — through weather, through night, continuously. Glaser proposed this in 1968; the NASA/DOE Reference System worked it out in full by 1979. It died on launch cost, not physics. Caltech's MAPLE demonstrator closed the loop end-to-end from orbit in 2023.

What this buys, in both cases: power becomes a property of a space rather than a property of a link. Storage stops being sized for the interval between charges and shrinks to the interval between harvests. The connector — the most common point of mechanical failure in any device — disappears as a design constraint. Energy reaches sealed, moving, and remote places copper cannot economically follow.

None of this requires new physics. Every element has been demonstrated at some scale. The reason it is not yet infrastructure is the subject of the next section.

Section 2 — The Bottlenecks

Every constraint below is decades old and non-negotiable. They explain why fifty years of demonstrations have produced almost no deployed infrastructure.

Inverse-square spreading. Power density falls with the square of distance. A 10 W transmitter delivers roughly 76 μ W to a small antenna at 5 metres. Filling a room to the legal exposure limit requires kilowatt-class radiated power for milliwatt-class delivery per device.

Diffraction. Beam width cannot fall below λ/D for aperture D . A 1 km transmitter at 2.45 GHz produces a beam whose first null spans roughly 10 km by the time it reaches the

ground from geostationary orbit — which is why the 1979 Reference System specified a 10×13 km rectenna. No array design escapes this.

Path loss. Atmospheric absorption is manageable at 2.45/5.8 GHz but rises sharply above 10 GHz, capping usable frequency. Ionospheric nonlinearity imposes a hard ceiling on beam intensity for space-to-ground links, independent of anything on the ground. Indoor multipath and polarisation mismatch make delivered power vary by an order of magnitude across a room.

Rectifier collapse at low density. Conversion efficiency exceeds 80 percent at high incident power and falls to 10–30 percent below $100 \mu\text{W}/\text{cm}^2$, with a hard floor near -20 dBm where diode forward voltage is never reached. Ambient harvesting operates precisely in the regime where rectifiers perform worst.

Multiplicative losses. DC-RF conversion, beam formation, collection, rectification, and conditioning each cost 5–30 percent. Multiplied through, a well-engineered high-power link tops out near 40–55 percent end to end. Diffuse ambient systems run in the fractions of a percent.

Exposure limits. ICNIRP and IEEE cap general-public exposure at roughly $10 \text{ W}/\text{m}^2$ ($1 \text{ mW}/\text{cm}^2$), averaged over 30 minutes. This is the binding constraint on the entire terrestrial architecture — a legally compliant room delivers, at best, low milliwatts per device, and any high-power mobile application is foreclosed outright: continuous kilowatt delivery to a moving target in public space is a directed-energy weapon by any reasonable definition, not an engineering challenge.

Regulatory vacuum. 2.45 GHz is unlicensed spectrum shared with Wi-Fi and Bluetooth; sustained kilowatt emission into it has no regulatory precedent. Space-to-ground transmission has no ITU allocation, no liability regime, and no cross-border framework.

Launch mass and specific power. The 1979 design massed 50,000 tonnes for 5 GW delivered. Reusable launch has cut cost-per-kilogram by roughly an order of magnitude — not yet enough. Specific power on orbit sits near $100\text{--}200 \text{ W}/\text{kg}$ against the roughly $1 \text{ kW}/\text{kg}$ viable economics require.

Six of these eight constraints are immovable by any amount of transmitter engineering — they are geometry, atmosphere, or biology. The two that are actually in motion, rectifier efficiency and launch economics, are where the rest of this thesis goes.

Section 3 — Design Philosophy

Sixty years of wireless power research optimized the wrong side of the equation.

The field decided early that the transmitter was the variable and the device was a fixed demand. Everything that followed is downstream of that single, catastrophic choice. If the device needs what it needs, the only remaining lever is to deliver energy more precisely. So the field built tighter beams, larger apertures, faster tracking loops, and spent careers shaving beam collection efficiency from 88 percent to 91. Every major programme is a variation on the same idea. Solar power satellite architectures are retrodirective by default:

the ground station emits a pilot, an orbital array of several billion elements phase-conjugates it, and the whole structure continuously corrects against its own thermal deformation to hold lock on a ten-kilometre rectenna. Terrestrial systems copy the pattern at room scale. The receiver announces its location. The transmitter aims.

This approach is a failure of imagination and a failure of taste.

Look at what shipping products already do. Ossia's Cota has devices pinging the transmitter up to a hundred times per second so power can be beamed back along the exact path the ping arrived on. For that to work, the transmitter maintains a live model of the room—the geometry, the reflectivity of every surface, and the position of every person standing in it—so the beam can route around bodies on its way to your pocket. Someone has already patented the obvious extension: derive the user's position from the trajectory of the power beam more accurately than any conventional indoor positioning system, then use that position to gate what the device is allowed to do. That was written up as a feature.

This is the world the architecture delivers. You walk across your living room and a beam walks with you, re-aiming a hundred times a second, because the infrastructure cannot give you anything until it knows exactly where you are. Your devices broadcast their coordinates continuously to a machine on the ceiling that has built a floor plan of your home and can tell you apart from your furniture.

This is aesthetically cringe. It is intrusive and invasive by construction. It is surveillance with a utility bill attached. It takes the one technology that should feel like abundance and converts it into continuous location tracking, then congratulates itself for the efficiency number on the chart. The lived texture of the thing is traded away to satisfy a metric that only matters under an assumption nobody ever re-examined.

The orbital version is the same crime at planetary scale. A one-kilometre transmitting aperture in geostationary orbit aimed at a ten-by-thirteen-kilometre field of rectifying antennas, fenced and excluded, exists for one purpose: to force diffraction back into a box after physics already told us it does not fit. A hundred and thirty square kilometres of ground hardware built to catch a beam that refuses to stay narrow. It is utilitarian in the ugliest sense of the word. It is a civilization straining against a constraint instead of designing something that lives comfortably inside it.

Point-to-point delivery is the architecture of scarcity. It is correct only if the receiving side needs a large, specific quantity of power at a specific place. Under that condition every joule is worth chasing, tracking earns its complexity, and the tenth device in a room genuinely costs the infrastructure something, because the infrastructure has to find it, aim at it, and hold the lock.

Six of the eight binding constraints—spreading, diffraction, atmospheric absorption, ionospheric nonlinearity, exposure limits, multipath—are geometry, atmosphere, and biology. None of them move. The field committed its entire research budget to pushing against the immovable half of the problem and treated the movable half as somebody else's department.

The movable half is device demand. It is the only term in the inequality that was ever free, and it is collapsing right now by orders of magnitude. Semiconductor scaling, microwatt-class sensing, and the migration of compute off the device into infrastructure that hyperscalers are already financing for reasons that have nothing to do with power delivery are doing the work. Once that number falls far enough, tracking has nothing left to justify it.

That inversion is the only interesting problem left. Everything else is cargo-cult engineering built on a premise that is already dying.

Section 4 — Let It Spread

My alternative is to stop fighting diffraction and start using it.

A transmitter radiates. The wave spreads, bends, reflects off walls, and fills the volume it is in. The field treats this as the enemy — the thing phased arrays exist to suppress, the reason apertures have to be kilometres wide. I treat it as the delivery mechanism. Saturate a space with low-intensity RF at the regulatory exposure limit, and every device inside that space harvests continuously from the field it is already sitting in. No pilot signal. No location reporting. No aiming. No handshake. No pairing, no docking, no charging behaviour of any kind. A device inside the coverage volume is powered, and a device outside it is not, and that is the entire user model.

This is how Wi-Fi works, and the parallel is exact at the architectural level. An access point does not negotiate a beam to your laptop. It radiates into a room, most of the energy lands on walls and floor and people, and the fraction that reaches your antenna is enough. Nobody has ever described Wi-Fi as wasteful. Coverage is the product; the discarded majority is the cost of never having to think about it.

The experience this produces is the reason I want it. Your device is powered continuously, from the moment you walk in, forever, with no cell inside it and no ritual attached to keeping it alive. No cable, no pad, no percentage in the corner of a screen, no planning your day around an outlet. The verb leaves the language. That is what abundant infrastructure is supposed to feel like — invisible, unmetered at the point of use, and taken entirely for granted.

Now the price, stated in full, because this architecture lives or dies on one number.

4.1 What abandoning the beam actually costs

Delivered power density is governed by Friis:

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2 \eta$$

Every term is fixed by the deployment except transmitter gain, G_t . Gain is the whole of what tracking buys, and giving it up is the whole of what this architecture pays.

A ceiling-mounted phased array with a 0.5 m aperture at 2.45 GHz achieves $G_t = 4\pi A/\lambda^2 \approx 211$, or 23 dBi. A transmitter shaped to flood a room downward operates around $G_t \approx 3$, or 5 dBi.

The cost of coverage is a factor of roughly 70 — about 18 dB of delivered power density, gone.

There is no engineering around this. It follows from the definition of gain. Concentration and coverage are the same quantity read in opposite directions.

The honest complement: indoor beam tracking never delivered the gain its advocates imply. A 0.5 m aperture at 2.45 GHz focuses, at 3 m range, to a spot of diameter $2.44\lambda d/D \approx 1.8$ m. The tracked beam indoors is already a body-sized blob. Sixty years of tracking machinery, a live 3D model of your home, and a hundred position pings per second buy 18 dB and a focal spot wider than the person standing in it.

4.2 The arithmetic of a saturated room

Fix the ceiling first. The FCC caps general-public maximum permissible exposure at **1 mW/cm² — 10 W/m² — above 1.5 GHz**, and ICNIRP and IEEE C95.1 land in the same place across the 2–6 GHz band. That is the number, and this architecture is designed to sit at it.

Holding 10 W/m² across a room requires $P_t = S \cdot 4\pi d^2 / G_t$. At $d = 5$ m with $G_t = 3$: **roughly 1.05 kW radiated per room.**

That number deserves to be stated plainly, because it is where the Wi-Fi analogy breaks. A Wi-Fi access point radiates 0.1 W. An ambient power layer at the regulatory ceiling radiates ten thousand times more. The analogy holds for topology and fails completely for energetics.

On the receiving side, effective aperture is $A_e = G_r \lambda^2 / 4\pi$ — set by wavelength, not by device size:

Frequency	λ	A_e (G=2)	Intercepted @ 10 W/m²	DC after rectification
900 MHz	33.3 cm	177 cm²	177 mW	~90-125 mW
2.45 GHz	12.2 cm	24 cm²	24 mW	~12-17 mW
5.8 GHz	5.2 cm	4.3 cm²	4.3 mW	~2-3 mW

Lower frequency collects an order of magnitude more per element, which is why every serious ambient system converges on the sub-gigahertz bands.

Three corrections keep this honest.

Rectification improves at this density. Incident power of 10 W/m² is 1 mW/cm², which is the regime where Schottky rectennas run 50–70 percent rather than the 30 percent they manage in ambient scavenging. Operating at the regulatory ceiling buys efficiency as well as raw power, and the gain compounds. At the edge of a coverage volume, where density falls to 1 W/m², efficiency drops back to 30–50 percent and harvested power falls by roughly a factor of twelve.

The effective aperture in that table is available only to an antenna of roughly $\lambda/2$ physical scale — 16 cm at 900 MHz. Shrink the structure below that and the Chu-Harrington limit takes the efficiency back. The receiving antenna, not the chip, sets the harvest.

And 10 W/m² is the legal maximum for continuous general-public exposure, which means a room designed to this figure holds its occupants at the regulatory ceiling permanently. The limit is defensible and internationally harmonised, and designing to it is compliant. It also leaves zero margin, and any deployment that wants headroom pays for it linearly in delivered power.

4.3 Where the inequality stands today

The condition for battery-free operation is one line:

$$P_{\text{harvested}} \geq P_{\text{device, average}}$$

Against an ambient field at the FCC ceiling — sub-gigahertz, 10 W/m², $\lambda/2$ -scale antenna — the left side delivers **roughly 8 mW at the edge of coverage and 90-125 mW at the design point.**

Now the right side:

Device class	Average draw	Status at 10 W/m ²
Duty-cycled environmental sensor	10-100 μ W	Closes with 3 orders of margin
Asset tag, e-ink label	50-500 μ W	Closes with 2 orders of margin
Hearable, low-rate wearable	5-30 mW	Closes
Smartwatch	20-50 mW	Closes at the design point
AR glasses (current architecture)	2-5 W	Misses by ~25×
Smartphone	1-2 W	Misses by ~12×

At the regulatory ceiling the picture inverts. Every low-power device class closes outright, and the wearable tier — hearables and watches, which the field has always treated as years away — closes today on physics that already exists. What remains open is the top tier: glasses and phones, missing by roughly one order of magnitude rather than the three the 1 W/m² case implied.

One order of magnitude is a gap that gets closed by the receiving side within a normal product cycle. Sixty years of RF engineering read that gap and concluded the field must get stronger. I read it and conclude the devices must get smaller in their demands — and the reason I am confident is that the entire semiconductor and data-centre industry is already closing it, at a scale no wireless power programme could ever fund, for reasons that have nothing to do with power delivery.

That is Section 5.

Section 5 — Two Kinds of Device

I want to ask a question that sounds too simple to be worth asking.

Why does a device need power at all?

Not how much. Not how efficiently. Why. Strip away the product category, the brand, the form factor, the roadmap, and ask what the electricity is actually being spent on. There are only two answers. Everything in this thesis depends on them being the only two.

A device spends energy to move information — capture it, process it, transmit it, or present it to a person. Or it spends energy to move matter — its own body, a payload, a joint, a wheel, a limb.

That is the complete list. Every function in every device belongs to one of those two, and the two obey completely different physics.

5.1 The question experts skip: does this have to happen here?

Before splitting by what a function costs, split by where it must occur. This is the test almost nobody applies, because device architecture is inherited rather than derived.

Take a pair of AR glasses apart and list what it does. Cameras. Inertial sensors. Microphones. A radio. A processor running perception, mapping, and rendering. A display. Now ask each one whether it has to be on the face.

The cameras do. The photons arrive at that position and nowhere else. The microphones do. The inertial sensors do — they measure the motion of that specific head. The display does, because the light has to enter that pupil. The radio does, because something has to talk.

The processor does not.

Bits have no location. A matrix multiply produces the same result in a wearable, in a cabinet down the street, or in a data centre in another state. There is no physical law binding computation to the place the data was captured. The only thing binding it is latency, and latency is a distance problem with a distance solution.

So the dumb question has a sharp answer: the single most power-hungry function in a modern wearable is also the only one with no reason to be there. Sensing, display, and radio are locked in place by physics. Compute is locked in place by habit.

That is the whole of the "compute leaves the device" argument. It requires no breakthrough. It requires noticing that one component was never fastened down.

5.2 Class A: devices that move information

Everything that survives the locality test — sensors, radio, display, and the minimum logic to run them — forms the first class. Glasses, hearables, watches, tags, badges, sensors, cameras, and every wearable form factor that has not been invented yet.

The cost structure of this class is bounded from below by Landauer's limit: erasing one bit at room temperature costs about 3×10^{-21} joules. Real silicon operates somewhere between three and six orders of magnitude above that floor, which means the information

side of computing has thousands to millions of times of theoretical headroom remaining. It has been descending that headroom for fifty years and continues to.

Once compute is relocated, something unexpected becomes the dominant load. The display does. Getting usable brightness into a pupil through a waveguide that throws away most of the light costs hundreds of milliwatts today, and that cost is set by optical efficiency and human photoreceptors rather than by transistor scaling. When compute leaves, the display becomes the floor. Anyone building toward a battery-free wearable should be spending their attention on waveguide efficiency and emitter physics. Almost nobody is.

5.3 Class B: devices that move matter

The second class does work against a physical field. Humanoids, mobile robots, manipulators, drones, actuated anything.

Lifting one kilogram by one metre costs 9.81 joules. That number was the same in 1600 and will be the same in 2600. There is no Landauer headroom here, no scaling curve, no process node. Motors already convert electricity to torque at better than 90 percent efficiency, which means the entire remaining optimisation space on the conversion side is a single-digit percentage.

A second dumb question, which is the one that matters most: on level ground, what is the theoretical cost of walking?

Zero. A passive dynamic walker descends a shallow slope using nothing but gravity and its own geometry, indefinitely, with no motors. Locomotion on flat ground is a cycle of falling and catching, and the energy released in the fall is available to fund the catch. Nothing in physics requires a walking machine to burn energy the way current ones do.

Real machines burn it anyway, because they brute-force every joint with an oversized motor and dissipate the recoverable energy as heat in the brakes. A humanoid today draws several hundred watts to over a kilowatt while walking. A human of the same mass spends a fraction of that. The gap between them is engineering waste rather than physical necessity, and it is the largest unclaimed efficiency in robotics.

So Class B splits again. The irreducible part is real work done against gravity — lifting, climbing, accelerating a payload. The recoverable part is the cost of locomotion itself, which current designs pay in full and should not.

5.4 The asymmetry

Set the two classes side by side and the shape of the problem appears.

	Class A: moves information	Class B: moves matter
Governing limit	Landauer, $kT \ln 2$	Newtonian work, mgh
Distance above the limit	3–6 orders of magnitude	Motors already >90% efficient
Direction of travel	Falling, fast, for fifty years	Flat, permanently
Can the function be relocated?	Compute yes; sensing, display, radio no	No — the mass is here
Typical draw today	5 mW – 5 W	300 W – 2 kW

Class A has enormous headroom and an industry spending hundreds of billions of dollars descending into it for unrelated reasons. Class B has essentially none on the conversion side, and its only real lever is refusing to waste energy it could have recovered.

Now measure both against the field from Section 4 — a room at the FCC ceiling delivering roughly 90 to 125 mW to a sub-gigahertz receiver:

Class A is one order of magnitude away, and the gap is closing on its own.

Class B is three to four orders away, and no amount of semiconductor progress touches it.

5.5 The recursion nobody talks about

One more question that sounds naive and is not: how much of a robot's energy is spent carrying the robot's own power supply?

A humanoid with a 1.5 kWh pack carries roughly 6 kg of cells, 8 to 10 kg with housing and thermal management. On a 70 kg machine that is over ten percent of total mass, and the cost of moving a legged system scales with mass. Ten-plus percent of every joule is spent transporting the thing that stores the joules — before accounting for the compute stack it also carries and the cooling that stack demands.

The battery is a passenger that charges rent. Every improvement in energy density is partly consumed by the mass of the improvement itself, which is why endurance figures move so slowly despite genuine chemistry progress.

5.6 What follows

These two classes need different answers, and treating them as one product category is the mistake underneath every current design.

Class A is solved by relocation. Move compute off the body, leave the functions physics pins in place, and the demand curve walks down to meet an ambient field. Nothing more is required.

Class B cannot be solved by relocation, because gravity does not care where the inference happened. It needs an architecture built specifically around the fact that its load is enormous, intermittent, and irreducible.

Section 6 takes Class A to its conclusion. Class B gets its own answer later, and it is the more interesting of the two.

Section 6 — The Device This Is For

Today's wearables and phones are the wrong target. Writing wireless power around them is the equivalent of designing lithium-ion packs for 1980s mainframes. The power numbers never collapse far enough, which is exactly why every company — like Energous, Ossia and the rest — that tried distance charging of smartphones failed and retreated to microwatt sensors.

So I am not writing this for phones or for the glasses currently shipping. I am writing it for the thing that replaces the screen.

I believe 2D displays are temporary. Not improved — abandoned. When you watch something or join a call, you will not look at pixels arranged on a plane. You will stand inside a photorealistic environment rendered around you in real time, with parallax, occlusion, directional sound, and lighting coherent enough that your brain stops auditing it and simply accepts the room it is in.

Nothing like this exists today. That is the point. This class of device is being defined right now, before its architecture hardens, which makes it the only category where the power question is still open.

Everything after this section is about powering it.

Section 7 — The Architecture of a Rendered World

Before any of the architecture, I want to state the objective in one line, because my earlier writing on this buried it under four domains and lost people.

Everything in this section exists to drive the power draw of the head-worn device low enough that a room can supply it.

That is the only goal. Compute leaving the device, nodes in buildings, choosing what to send over the air — each is a method for removing work from something that has no battery. Every milliwatt removed is a milliwatt the field no longer has to deliver. When someone asks why any of this is worth building, that is the answer, and there is not a second one.

Section 4 gave the number. A room at the FCC ceiling delivers roughly 100 mW to a sub-gigahertz receiver. So the spec for the entire immersive stack is: **build a device that stands inside a rendered world on 100 milliwatts.**

Now design backwards from that.

7.1 The budget, itemised

Here is what a device delivering full spatial presence has to run, and what each piece costs today:

Function	Draw
Cameras (duty-cycled or event-driven)	5–20 mW
Inertial sensors	0.5–3 mW
Microphones	1–2 mW
Eye tracking	5–10 mW
Minimal local logic	5–20 mW
Radio	10 mW – 1 W
Display	50–300 mW
Perception, mapping, rendering	2–5 W

The first five lines add to under 60 mW. They fit. They have fit for years.

The last three are the entire problem, and they fail in different ways. Rendering fails by a factor of thirty. Radio fails or passes depending on a decision nobody has made yet. Display fails by a factor of two or three and is a physics problem.

Three problems, three answers.

7.2 Rendering: the easy one

Rendering is the largest line in the table and the only one with no reason to be on the head. Section 5 established why — bits have no location. A frame rendered in a cabinet down the street is identical to one rendered against someone's forehead, except that the cabinet is plugged into a wall and can be cooled with a fan.

Removing rendering and perception from the device cuts its draw by roughly 95 percent. Nothing else on the list comes close to that. There is no silicon breakthrough required and no new physics involved. It is a decision about where a chip sits, and the wearable industry made that decision wrong by copying the smartphone without asking why.

That single move takes the device from watts to hundreds of milliwatts. It is the difference between needing a battery and not.

7.3 Radio: the decision that decides everything

Once rendering moves off, something has to cross the air, and the radio bill is set by one thing: **how many bits**. A good short-range radio costs roughly 1 to 10 nanojoules per bit.

Multiply that out:

What you send	Data rate	Radio power
Raw volumetric / light field	1-10 Gbps	1-10 W — dead on arrival
Compressed stereo video	100-500 Mbps	100-500 mW — busts the budget
Semantic description + cached scene	1-10 Mbps	1-10 mW — fits with room to spare

So the question of what format the spatial internet uses is a power question wearing a costume. Send pixels and the radio alone consumes more than the entire harvest. Send a description of the scene and let the receiver rebuild it, and the radio nearly disappears — while the rebuilding cost lands right back on the device you were trying to keep empty.

Both extremes fail. The rule I would build to:

Send a thing only when sending it costs the device less energy than recalling or reconstructing it.

A wall that has not moved in an hour is already in memory. Recall it. A texture the device has seen a thousand times is a pointer, not a payload. A face mid-expression gets sent at full fidelity, because generating a face is expensive and untrustworthy. Every object, every frame, gets that decision made for it against one budget.

Nobody has built this layer. It matters more than any codec, because it is the difference between a radio that fits inside 100 mW and one that does not.

7.4 Where the compute goes, and why distance is a power problem

Spatial presence breaks if motion-to-photon latency exceeds roughly 20 milliseconds. Spend that budget: sensor readout 2-5 ms, uplink 1-2 ms, inference and render 5-8 ms, downlink 1-2 ms, display scanout 2-5 ms. Almost nothing is left for travel time.

Light in fibre covers about 200,000 km per second, so a 1,000 km round trip costs 10 ms before a single packet is switched or queued. Hyperscale data centres are built where land and power are cheap, which is a long way from where people live. The distance does not negotiate.

Here is the part that connects it back. **A node that is too far away forces the device to compensate locally** — predicting head motion, warping the last frame, holding a local scene cache, patching what arrived late. Every one of those is compute back on the head, and compute on the head is the thing we just removed. Latency failure converts directly into power failure.

So the compute goes into cabinets a few kilometres out, or into the building itself. Not for elegance. Because that is the only placement where the device gets to stay empty.

Sharing makes it affordable. A wearable's GPU sits idle most of the day and is sized for its worst second. A neighbourhood node runs near full utilisation, because a hundred people's

activity averages out. Same silicon, a fraction of the cost per person, and cooling handled by a fan rather than by someone's forehead.

7.5 The room becomes the computer

Push it one step further and the building holds the compute directly, alongside its electrical, network, and HVAC plans. The reason is mundane and decisive: **the building is already plugged in**. Grid power at a fixed location is effectively unlimited, thermally unconstrained, and costs nothing to carry.

A room that senses its own geometry and holds its own persistent model gives every device inside it a second discount. The device stops re-deriving what the room already knows. Fewer bits over the radio, less sensing duty cycle, less local logic. The building does the work once for everyone in it.

7.6 What is left

Run the three moves and the budget closes:

- Rendering and perception offloaded: **2-5 W → under 50 mW**
- Adaptive representation over the radio: **500 mW → 5 mW**
- Sensors, unchanged: **~30 mW**

Total: under 90 mW, inside the field a room can legally supply.

One line refuses to move. The display still costs 50 to 300 mW, and no amount of architecture touches it, because the cost is set by waveguide efficiency and human photoreceptors. **Once compute leaves the device, the display is the entire remaining problem**. Everything else in this thesis is solved by placement. That one is solved by optics, and it is where I would put the next decade of attention.

Class A closes here. Section 8 takes on Class B, where none of these moves work, because gravity does not care where the inference happened.

Section 8 — The Class B Problem

Section 5 drew the line. Class A moves information and closes against an ambient field within one order of magnitude. Class B moves matter, and I put the gap at three to four orders of magnitude. This section earns that number.

Take Tesla Optimus, because it is the most publicly documented humanoid on earth and there is no reason to argue from a hypothetical when a real machine gives better numbers.

8.1 The numbers, stated plainly

State	Power draw
Idle, standing	~100 W
Walking, normal pace	250–500 W
Sustained heavy manipulation	500 W – 1 kW
Single arm, active lift	1–2 kW
Joint acceleration transient (reported)	up to ~3 kW
Onboard energy storage	2.3 kWh
Runtime at typical load	4–8 hours

Every figure in that table is publicly reported. None of it is my extrapolation.

Now put it next to the field. Section 4 gave a saturated room at the FCC ceiling: roughly 90 to 125 mW delivered to a well-designed sub-gigahertz receiver.

Idle Optimus needs 100 W. The room delivers 0.1 W. That is a gap of three orders of magnitude before the robot has taken a single step.

Walking pushes the requirement to 250–500 W — four orders of magnitude past the field. A single arm mid-lift draws more power on its own than a fully saturated room delivers to a thousand devices combined. The transient spike during joint acceleration hits 3 kW, which is thirty thousand times what the room supplies.

8.2 Why the Class A playbook does not apply

Relocating compute does nothing here, because the power is not going to inference. It is going to torque. A humanoid's perception and planning stack draws perhaps 50 to 150 watts, and every watt of it can leave the robot the same way it leaves a pair of glasses. That solves maybe 10 percent of the budget. The other 90 percent is motors converting electricity into force against a body gravity is actively pulling down, and gravity does not care where the inference happened.

Efficiency gains do not close a three-order gap either. Motors already convert electricity to mechanical work above 90 percent efficiency. There is no headroom left to shave.

And a bigger beam cannot legally exist. Section 2 already fixed the ceiling at 10 W/m². Closing this gap by raising transmitted power means delivering kilowatts continuously to a moving target in occupied space — which Section 3 already named for what it is: not infrastructure, a directed-energy weapon pointed down a hallway. That option is off the table, not merely disadvantaged.

Class B needs its own architecture. That is Section 9.

8.3 The loop nobody escapes

The robotics industry has a circular problem and has been treating it as a linear one.

A humanoid draws power. Peaks are brutal, so endurance is short. The response is a bigger battery. A bigger battery is more mass. More mass means more torque at every hip and knee, on every step, forever — which means more power draw, which means the bigger battery buys less runtime than its capacity suggested, which motivates a bigger battery.

Every iteration of that loop makes the machine heavier and the return smaller. I want to put actual numbers on it, because the loop has a mathematical shape and the shape has a wall in it that I have not seen stated anywhere.

The mass audit. Optimus Gen 2 weighs 57 kg and carries a 2.3 kWh pack. At realistic pack-level energy density of around 170 Wh/kg — cells plus housing, BMS, and thermal management — that pack is roughly **13.5 kg. Twenty-four percent of the entire robot.**

Nearly a quarter of the machine exists to feed the machine.

The loop, written down. Locomotion power for a legged system scales roughly with mass. Call that constant k ; for Optimus at ~375 W average walking draw and 57 kg, $k \approx 6.6$ W/kg. Let ρ be pack energy density and t the runtime you want. Then:

$$m = m_{\text{structure}} + \frac{k \cdot m \cdot t}{\rho} \Rightarrow m = \frac{m_{\text{structure}}}{1 - kt/\rho}$$

The battery mass depends on total mass, and total mass depends on battery mass. This is a rocket equation wearing a different costume, and it behaves like one.

Run it for Optimus, structure mass 43.5 kg:

Target runtime	Total mass	Battery mass	Battery share
6 hours	57 kg	13.5 kg	24%
12 hours	82 kg	38 kg	47%
18 hours	175 kg	131 kg	75%
24 hours	640 kg	596 kg	93%

And there is a wall. The denominator hits zero at $t = \rho/k$. For these numbers that is **25.8 hours**. No battery-powered humanoid at current energy density and current locomotion efficiency reaches 26 hours of walking endurance, regardless of how large you build the pack. Past that point the battery is carrying only itself. Adding capacity subtracts runtime.

That wall does not move with better cells the way people assume. Doubling energy density to 340 Wh/kg pushes it to 51 hours — a real improvement, and still a wall, and still bought with thirty years of chemistry that has delivered roughly a tripling in as long.

Now the dumb question: what is this robot if you take the battery out?

Delete the pack and Optimus is a **43.5 kg machine**. Apply Section 7's argument and offload the perception and planning stack — the compute board, its heatsink, its cooling — and you remove another 2 to 4 kg. Call it **40 kg**.

Recompute the power draw at 40 kg instead of 57 kg:

	With battery (57 kg)	Without (40 kg)
Idle, standing	100 W	~70 W
Walking	375 W	~264 W
Arm lift	1–2 kW	0.7–1.4 kW

The 40 kg figure assumes compute has been offloaded to infrastructure per Section 7. The 264 W is purely actuation power with no on-board inference.

Thirty percent off every number in the machine's power budget, from removing components rather than improving any of them.

That is the figure wireless power should be designed against. Every published analysis of powering a humanoid wirelessly starts from 57 kg and 375 W and concludes it is impossible. It starts from the wrong machine. The robot you would actually build for an ambient field never had the pack in it, and it is 30 percent cheaper to move as a direct consequence.

And the loop reverses direction. In the battery architecture, every kilogram of storage adds power draw which demands more storage — the recursion diverges. Take the storage out and the recursion runs backwards: less mass means less power, which means less energy to supply, which means less onboard hardware, which means less mass. The same feedback that made batteries diverge makes a battery-free machine converge.

8.4 What a human costs

I want to put Optimus next to the machine it was copied from.

A human basal metabolic rate is roughly **100 watts** — about 2,000 kcal a day, spread across 1.7 m² of skin at around 60 W/m². That number is not spent on standing. Standing is free. That 100 watts buys breathing, blood circulation to every cell in the body, immune surveillance, continuous tissue repair, digestion, and holding a 37 °C core temperature against whatever the room is doing. Roughly 70 percent of it goes to organ and cellular function, about 20 percent to physical activity, and around 10 percent to thermoregulation specifically.

And inside that same 100 watts, **the brain runs on about 20.**

Twenty watts of general intelligence. Vision, language, planning, motor control, memory, and abstraction, on less power than a household light bulb dims to.

Now set the two side by side:

Human (~70 kg)		Optimus (57 kg)
Baseline draw	~100 W	~100 W
What that buys	Circulation, digestion, immunity, repair, thermoregulation, and a 20 W general intelligence	Holding a pose
Cognition	~20 W	50–150 W, narrow
Walking	~300–400 W metabolic	~375 W

The two machines draw the same power at rest. One of them is running an entire biology on it. The other is standing still.

This is the comparison that should embarrass the field, and I mean that constructively. **A robot pays none of biology's overhead — no digestion, no immune system, no cellular repair, no blood to move, no core temperature to defend — and still burns as much as a living body doing all of it simultaneously.** Its cognition alone costs three to seven times what a human brain costs while doing far less.

So when someone tells me a humanoid's power draw is near some physical floor, I do not believe them. There is a 70 kg proof walking around that the floor is much lower. The gap between Optimus and a human is engineering debt, and debt gets paid down.

8.5 Three variables, one of them locked

Back to the equation everything in this thesis runs on:

$$P_{\text{harvested}} = S \times A \times \eta$$

Incident power density, effective collecting area, conversion efficiency. Three terms. Take them one at a time and see which ones a designer actually controls.

***S* is locked.** The FCC caps it at 10 W/m². Sections 2 and 3 settled this — the limit is biological, internationally harmonised, and not a lobbying target. Treat it as a constant and design underneath it.

***A* is where robots have an advantage nobody uses.** In Section 8.1 I quoted a three-order-of-magnitude gap. That number came from a wearable's antenna — 177 cm² of effective aperture. A humanoid is not a pair of glasses. It has roughly **2 m² of surface**, the same order as human skin, and every square centimetre of it is a candidate receiving surface if the skin itself is the rectenna.

Two square metres against 177 square centimetres is a factor of **113**. The robot's own size, which is a liability in every other part of this analysis, is an asset here.

Rerun the harvest with the robot's own body as the aperture, at 10 W/m² and 60 percent rectification:

Condition	Harvested
Fully illuminated, 2 m ² presented	~12 W
Orientation-averaged, ~0.7 m ² effective	~4 W

The honest number is the second one. Effective area is the *projected* area facing the field, and it changes continuously as the machine turns, bends, reaches, and walks. Anyone quoting 2 m² for a moving robot is quoting a best case. Real deployment needs multi-directional transmitters and receiving surfaces distributed so some usable fraction stays illuminated in any pose.

Conformal metamaterial rectennas are what make this term real rather than theoretical — curved, structural, integrated into the shell instead of bolted on as a panel. This is an open research problem and I am not going to pretend otherwise. It is also the single highest-leverage unsolved thing in Class B, because it is the only term with two orders of magnitude sitting in it.

η is the other live variable. Rectification runs 50–70 percent at the densities this architecture operates in, and it improves for reasons unrelated to robotics.

So: one term fixed by law, one term with enormous unclaimed headroom, one term improving on its own. Which leaves the fourth quantity, the one on the other side of the inequality.

P_{required} . Section 8.3 already took 30 percent off it by deleting the battery and the compute stack. Section 8.4 established there is far more to take, because biology does the same job on a fraction of the budget.

Against a 40 kg battery-free machine drawing ~70 W idle and ~264 W walking, harvesting ~4 W orientation-averaged, the remaining gap is **roughly one order of magnitude**. That is the real number, and it is a different problem entirely from the three orders I opened this section with.

8.6 Trickle-spike

Here is the observation the whole architecture turns on: **a robot's load is not flat, and the shape of it is generous.**

A humanoid spends the overwhelming majority of its duty cycle near baseline — standing, sensing, balancing, waiting, moving slowly. Call it 90 percent of the time. The remaining 10 percent is where the kilowatts live: lifting, catching, accelerating a joint, arresting a fall. Those events last fractions of a second.

Sizing a power system for those spikes is exactly what produces the battery brick. And the spikes do not need to be *supplied* by the field. They need to be **buffered**.

That is the trickle-spike architecture:

Harvest the baseline continuously from the ambient field. Buffer the spikes locally in onboard storage.

Supercapacitors are built for this job — high power density, fast discharge, hundreds of thousands of cycles, and complete indifference to the deep-discharge behaviour that destroys chemical cells. Paired with a small solid-state cell for reserve, they absorb the peak and refill from the trickle during the far longer intervals between peaks.

The energy balance is one line: over any duty cycle, harvested energy must meet or exceed consumed energy. Storage decouples instantaneous supply from instantaneous demand, which means the machine can momentarily draw far more than the field delivers, provided the average closes.

This is the reframing that matters. Onboard storage stops being sized for endurance and becomes sized for the largest single burst. Endurance comes from the field. Storage handles transients. Those are different components, orders of magnitude apart in capacity, and conflating them is what put a 13.5 kg pack on every humanoid in development.

This way we can achieve the best of both worlds — wireless power dream without the danger of death ray and handling high power requirements without carrying bricks.

8.7 Why the gap is worth closing

I should be direct about what this paper is and is not.

This paper does not solve Class B. The gap is around one order of magnitude, and closing it needs conformal rectenna surfaces that do not exist yet, actuation that wastes far less than current designs do, and orientation-averaged collection measured in real environments rather than modelled on paper. None of those are things I can hand anyone. The first-principles questions each one raises are collected in Section 10.

What this paper asserts is that the gap is one order of magnitude rather than three, that every term in it except the safety ceiling is moving in a favourable direction, and that **ambient-powered robotics is a systems design problem rather than a physics problem.** That distinction is the entire argument, and it decides whether serious people spend their careers on it.

Here is why they should.

A human works roughly 2,080 hours a year. A robot that never stops to charge, at 95 percent uptime, works 8,322. That is a **4x multiplier on labour hours**, and it comes from removing the charging cycle alone — 3x from the shift arithmetic, the rest from holidays, breaks, and shift changeover that a machine tied to no cell simply does not have.

Today's humanoids run four to eight hours and then go to a dock. Every one of them is a machine that works one shift and rests two, and the resting is imposed by a component the machine did not need to carry in the first place.

Remove it, and physical labour stops being metered in shifts. That is what abundance actually looks like when you trace it back to a mechanism: continuous, uninterrupted physical work at a marginal cost that approaches the cost of the electricity, available to anyone, everywhere the field reaches.

That is worth an order of magnitude of somebody's attention.

Section 10 — Research Agenda

I have tried, through this whole paper, to ask the questions an expert stops asking after enough years in a field — the ones that sound too basic to raise in a room full of people with PhDs, which is exactly why they never get answered. This section collects them. None have settled answers. I am not going to pretend otherwise, and I am not going to dress up a guess as a finding anywhere below. Some of these are dissertations. A few are companies. One or two might be the whole thing.

On the transmitter side, where I made you accept a ceiling

If 10 W/m^2 is fixed by biology and not by engineering, what does a room actually look like once every surface — ceiling, wall, fixture — participates in holding that density instead of relying on one point source? Does distributing the same total power across many low-intensity emitters change the exposure calculus, or only the aesthetics of the installation?

What does occupied, furnished, human-filled multipath do to delivered density in practice, versus the free-space numbers I used throughout Section 4? I ran the arithmetic in a vacuum. Nobody has published what a living room with three people and a couch actually measures.

On the receiving side, where the whole architecture lives or dies

Section 4 fixed effective aperture at $\lambda/2$ physical scale. Is that actually a hard floor, or is it a rule of thumb inherited from antennas designed for something other than harvesting? What happens at the Chu-Harrington limit specifically for a rectifying receiver rather than a transmitting one?

Section 8 leaned an entire order-of-magnitude improvement on conformal metamaterial rectennas covering a robot's own skin. Nobody has published an orientation-averaged efficiency number for a curved, multi-surface receiver moving through a realistic gait cycle. Until that number exists, my 4-watt figure is a calculation, not a measurement, and I want to be honest that the difference matters.

I used 50–70 percent rectification throughout this paper as the working figure at 10 W/m^2 , and I never asked whether that is close to a ceiling or whether real headroom sits above it. Goldstone hit over 80 percent, but at far higher incident density than the regulatory limit allows indoors. What is the actual maximum achievable rectenna efficiency specifically at 10 W/m^2 — is 70 percent close to the physical limit for a diode-based rectifier at this density, or is that number an artifact of circuits nobody has optimised for this specific operating point because nobody needed to before now?

On the question I keep returning to and never fully answer

Section 8.4 put a human next to a robot and found the robot losing on baseline draw despite carrying none of biology's overhead. I do not have a complete answer for why. Is it the actuator topology — direct-drive motors fighting their own inertia where tendons and elastic elements would store and return energy the way muscle does? Is it the control architecture — continuous high-frequency correction where a nervous system uses sparse, event-driven signalling? Or is it simply that nobody has had to care yet, because grid power made the question uninteresting? I suspect it is mostly the third one, and if I am right, the fix is cultural before it is technical.

On the load itself

What is the true baseline draw of a humanoid with the battery and the compute stack both removed, measured rather than derived the way I derived it in Section 8.3? My 40 kg, 70-watt figure is an extrapolation from published Optimus data. Someone needs to build the machine and put a meter on it.

What fraction of a robot's peak-to-baseline ratio is genuinely irreducible — real work against gravity — versus recoverable, the way Section 5.3's passive dynamic walker suggests most locomotion cost is? Nobody has published that split for a legged humanoid doing real tasks rather than walking on a flat lab floor.

On the floor itself, for both classes at once

Every number I used for device demand in this paper is a current-generation number — today's silicon, today's actuators, today's control loops. I never asked the sharper question: given the best actuation and the most power-efficient computing that physics actually permits, not what ships today, what is the true minimum average draw for a Class A device doing full spatial sensing and display, and for a Class B device doing useful physical work? Section 5 gave the governing limits — Landauer for one, Newtonian work for the other — but a governing limit is not a floor. Nobody has done the engineering work of asking how close either class can actually get to its limit, and until someone does, every gap I have quoted in this paper is a gap against today's engineering, not against what is truly achievable.

On storage, since trickle-spike only works if the spike side holds up

What supercapacitor energy density, at what cycle life, actually closes the gap for a real duty cycle of lifts, catches, and recoveries rather than the clean 90/10 split I used for illustration? I named the mechanism in Section 8.6. I did not size it against a real task distribution, because that distribution does not exist in the literature yet.

On the outer layer, which I have mentioned twice and developed nowhere

Terrestrial and orbital ambient power are the same architecture at different altitude, as I argued in the earlier paper this one grew out of. What receiver architecture actually serves both without being over-built for one and under-built for the other? I have not touched this seriously here, and I do not think it should be touched seriously until the terrestrial case closes first.

The one I think matters most

If Class A closes this decade — and Section 7's arithmetic says it does — and Class B closes within the following one, what changes about how a city is built? Not the power layer. The rest of it. Streets, buildings, labour, ownership, what a factory is, what a home is. I do not know the answer. I am not sure the question has been asked seriously by anyone who also understands the physics in Sections 2 through 8, and I think it should be.

Conclusion — Why I Wrote This

I have spent ten sections on glasses, rooms, and robots. None of them are the reason I wrote this paper.

Intelligence scales with energy. Not with data — we have enough, and can generate more. Not with clever training tricks — those buy percentage points, not orders of magnitude. By 2026 the industry arrived at this without my help: the binding constraint on frontier AI stopped being chip supply and became electricity. A single training campus now needs 100 megawatts to a gigawatt of dedicated power — a nuclear plant's output, aimed at one building. Hyperscalers are building their own plants on-site because grid queues are now the ceiling on how smart a model can get. That is the tracked-beam mistake from Section 3, replayed at civilizational scale — point-to-point, exclusive, one dedicated plant per campus.

My claim: the ceiling on energy capture is architectural, not physical. Physicists put a Kardashev Type II civilization a thousand to two thousand years out. That number is right about today's tools — human-paced construction, grid-bound, point-to-point. It assumes the next thousand years get built like the last thousand.

Here is the bridge. $P_{harvested} = S \times A \times \eta$ does not know whether it is powering glasses, a robot, or a data center. A room at the FCC ceiling and an orbital farm beaming gigawatts to a rectenna field are the same architecture at different apertures. Solve ambient and orbital delivery once, and you have not built a wearable feature. You have built the delivery mechanism for the one resource gating every further gain in machine intelligence.

Now the loop that collapses the timeline. Section 8.7 gave robots roughly four times human working hours once they stop needing a dock. Free of batteries, they do not just do warehouse work — they build rectenna farms and launch collectors twenty-four hours a day, bounded only by how much of the field they stand in. More energy trains more capable models. Those models get pointed at exactly the open problems in Section 10 — rectenna efficiency, actuation, launch cost — and close them faster than any human research community would. More energy makes more intelligence; more intelligence closes the energy problems faster. That loop does not run on institutional timescales.

I do not think we reach Type II in fifty to sixty years by getting marginally better at solar panels. I think we reach it by removing the assumption that energy must be delivered point-to-point and stored locally — the same assumption I spent this paper arguing against for a pair of glasses. This paper is the first piece of that argument.